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Entanglement : state that cannot be written as a product of single-particle states

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2 electrons in the Coulomb field of a nucleus of charge Ze.

Consider spin-independent forces, no relativisitc correction, and we treat the nucleus as heavy.

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} - \frac{Ze^2}{r_1} - \frac{Ze^2}{r_2} + \frac{e^2}{r_{12}}$$

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Ground state : $\frac{1}{\sqrt{2}}[1s(1)1s(2\rangle + 1s(2)1s(1)].\frac{1}{\sqrt{2}}(|\alpha\beta\rangle - |\beta\alpha\rangle)$

: only singlet is possible, $^1S_0$

$S = 0$	$S = 1$
singlet	triplet

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Excited states : 1s2s

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Ortho helium	Para helium
He atoms with symmetric (parallel) spin function $\|\alpha\alpha\rangle$, $\|\beta\beta\rangle$, $\frac{1}{\sqrt{2}}(\|\alpha\beta\rangle + \|\beta\alpha\rangle)$	He atoms with anti-symmetric (anti-parallel) spin function $\frac{1}{\sqrt{2}}(\|\alpha\beta\rangle - \|\beta\alpha\rangle)$

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metastable state, $^3S_1$	energetically the lower state, $^1S_0$

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$$\hat{h}(i)\ u_n(i) = \epsilon_n(i) u_n(i);$$

$$u(i) \equiv |n, l_i, m_i, m_s\rangle_i; \quad \epsilon_n = -\frac{2^2 e^2}{2a_0}\frac{1}{n^2} = -4 \times \frac{13.6}{n^2}\ \text{eV}$$

$\psi_{\mathsf{He}} = \phi(\vec{r_1}, \vec{r_2})\ \chi_0^0$: singlet, or $\phi(\vec{r_1}, \vec{r_2})\ \chi_{S=1}^{Ms}$: triplet

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zeroth order : electrons are non-interacting
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$\phi_s^{(0)} = \frac{1}{\sqrt{2}} [|n\, l\, m_l\rangle_1 |n'\, l'\, m_l\rangle_2 + |n\, l\, m_l\rangle_2 |n'\, l'\, m_l\rangle_1]$

: symmetric w. r.t. permutation operator $\hat{P}(1,2)$

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and $\phi_a^{(0)} = \frac{1}{\sqrt{2}}\left[|n\ l\ m_l\rangle_1 |n'\ l'\ m_l\rangle_2 - |n\ l\ m_l\rangle_2 |n'\ l'\ m_l\rangle_1\right]$: antisymmetric.

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$\psi_{\text{singlet}} = \phi_s^0\chi_0^0$;

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$\psi_{\text{singlet}} = \phi_s^0\chi_0^0$; similarly, $\psi_{\text{triplet}} = \phi_a^0\chi_1^{Ms}$

There is a large degeneracy here before we introduce H', i. e., interelectronic repulsion.

Consider $n = 1$, $n' = 1 \Longrightarrow l = 0$, $l' = 0 \Longrightarrow L = 0 + 0 = 0$

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Now consider $n = 1,\ n' = 2 \Longrightarrow l = 0,\ l' = 0$ (s) or 1 (p)

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For $L = 0$, we have singlet. $S = 0,\ J = 0$ and triplet, $S = 1;\ J = 1$

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: $^1S_0$ and $^3S_1$ respectively

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Now consider $n = 1,\ n' = 2 \Longrightarrow l = 0,\ l' = 0$ (s) or 1 (p)

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: $^1S_0$ and $^3S_1$ respectively

For $L = 1$, similarly, we will get singlet, $S = 0$, $J = 1$

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: $^1S_0$ and $^3S_1$ respectively

For $L = 1$, similarly, we will get singlet, $S = 0,\ J = 1$ and triplet, $S = 1;\ J = 2, 1, 0$

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Now consider $n = 1,\ n' = 2 \Longrightarrow l = 0,\ l' = 0$ (s) or 1 (p)

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: $^1S_0$ and $^3S_1$ respectively

For $L = 1$, similarly, we will get singlet, $S = 0,\ J = 1$ and triplet, $S = 1;\ J = 2, 1, 0$

: $^1P_1$ and $^3P_2, ^3P_1, ^3P_0$ respectively

There is a large degeneracy here before we introduce H', i. e., interelectronic repulsion.

Consider $n = 1,\ n' = 1 \Longrightarrow l = 0,\ l' = 0 \Longrightarrow L = 0 + 0 = 0$

$J = L + S = 0 + 0 = 0$

$\therefore$ this state is $^1S_0$

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Now consider $n = 1,\ n' = 2 \Longrightarrow l = 0,\ l' = 0$ (s) or 1 (p)

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For $L = 0$, we have singlet. $S = 0,\ J = 0$ and triplet, $S = 1;\ J = 1$

: $^1S_0$ and $^3S_1$ respectively

For $L=1$, similarly, we will get singlet, $S=0,\ J=1$ and triplet, $S=1;\ J=2, 1, 0$

: $^1P_1$ and $^3P_2, ^3P_1, ^3P_0$ respectively

Each of these has energy$= \epsilon_1 + \epsilon_2 = \left(-2^2 \times 13.6 - \frac{2^2}{4} \times 13.6\right)$ eV

$= -2^2(13.6)(1 + \frac{1}{4}) = -17 \times 2^2 = -68$ eV

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Finally for $L = 0$ and 2, spatial part is symmetric and spin singlet, no $^3S_1$

3 states gone

Unperturbed configuration

$n = 2,\ n' = 2$ $\overline{(2s2s)}$ $\overline{(2s2p)}$ $\overline{(2p2p)}$ $E^{(0)}_{2,2} = -27.2$ eV : ${}^1S_0, {}^1D_2, {}^3P_2, {}^3P_1, {}^3P_0$

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$n = 1,\ n' = 1$ $\overline{1s1s}$ $E^{(0)}_{1,1} = -108.8$ eV : ${}^1S_0$:

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$$\phi^0_{\text{ground}} = |n\, l\, m_l\rangle_1 |n'\, l'\, m_l\rangle_2 = \sqrt{\left(\frac{Z^3}{\pi a_0^3}\right)} e^{-\frac{Z}{a_0} r_1} \cdot \sqrt{\left(\frac{Z^3}{\pi a_0^3}\right)} e^{-\frac{Z}{a_0} r_2} = \left(\frac{Z^3}{\pi a_0}\right) e^{-\frac{Z}{a_0}(r_1 + r_2)}$$

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$$E^{(1)}_{\text{ground}} = \frac{e^2}{\pi^2}\left(\frac{Z^6}{a_0^6}\right) \int d\vec{r}_1 d\vec{r}_2 .\ \frac{1}{|\vec{r_1} - \vec{r_2}|} e^{-\frac{2Z}{a_0}(r_1 + r_2)} \quad \text{——————————(1)}$$

$$\frac{1}{|\vec{r_1}-\vec{r_2}|} = \frac{1}{\sqrt{r_1^2+r_2^2-2\vec{r_1}.\vec{r_2}}}$$

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$$\therefore \frac{1}{|\vec{r_1}-\vec{r_2}|} = \frac{1}{\sqrt{1+\left(\frac{r_<}{r_>}\right)^2-2\frac{r_<}{r_>}\hat{r_1}.\hat{r_2}}} = \frac{1}{r_>}\sum_{l=0}^{\infty}\left(\frac{r_<}{r_>}\right)^l P_l(\hat{r_1}.\hat{r_2})$$

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The integral in Eq. (1) turns out to be $\frac{5}{8}\pi^2 a_0^5$

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J is called the Coulomb integral and K the exchange integral.

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$$I_1 = 16\pi^2 \int_0^{\infty} r_1 dr_1 e^{-2\zeta r_1} \left[\int_0^{r_1} r_2^2 e^{-2\zeta r_2} dr_2 + \int_{r_1}^{\infty} r_2 e^{-2\zeta r_2} . r_1 dr_2\right]$$

$$= 16\pi^2 \int_0^{\infty} r_1 dr_1 e^{-2\zeta r_1} \left\{\left[\frac{e^{-2\zeta r_2}}{(2\zeta)^3}\left(-4\zeta^2 r_2^2 - 4\zeta r_2 - 2\right)\right]_0^{r_1} + \left[\frac{e^{-2\zeta r_2}}{(2\zeta)^2}\left(-2\zeta r_2 - 1\right) r_1\right]_{r_1}^{\infty}\right\}$$

$$= 16\pi^2 \int_0^{\infty} r_1 dr_1 e^{-2\zeta r_1} \left[e^{-2\zeta r_1}\left(\frac{-2\zeta r_1 - 2}{8\zeta^3}\right) + \frac{2}{8\zeta^3}\right]$$

$$\int_0^{\pi} \sin\theta_2 d\theta_2 \int_0^{2\pi} d\phi_2 \left(r_1^2 + r_2^2 - 2r_1 r_2 \cos\theta_2\right)^{-\frac{1}{2}} = 2\pi \left[\frac{(r_1+r_2)-|r_1-r_2|}{r_1 r_2}\right]$$

Putting this in the integral and carrying out the integration over the angular coordinates of electron 1:

$$I_1 = \int_0^{\infty} r_1^2 dr_1 \int_0^{\infty} r_2^2 dr_2 \int_0^{\pi} \sin\theta_1 d\theta_1 \int_0^{2\pi} d\phi_1 . e^{-2\zeta r_1} e^{-2\zeta r_2} . 2\pi \left[\frac{(r_1+r_2)-|r_1-r_2|}{r_1 r_2}\right]$$

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$$\Longrightarrow I = \frac{5}{8}\frac{Ze^2}{a_0}$$

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The theorem can be extended to higher energy states also, with a bit of modification

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$$E_{\text{trial}} = \frac{e^2}{a_0}\left\{-Z^{*^2} + 2Z^*(Z^* - Z) + \frac{5}{8}Z^*\right\}$$

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$$\langle \phi_{\text{tr}}|\tfrac{1}{r_1}|\phi_{\text{tr}}\rangle = \langle 1s'(1)|\tfrac{1}{r_1}|1s'(1)\rangle . \langle 1s'(2)|1s'(2)\rangle = \langle 1s'|\tfrac{1}{r}|1s'\rangle = \langle \phi_{\text{tr}}|\tfrac{1}{r_2}|\phi_{\text{tr}}\rangle$$

$$\therefore E_{\text{trial}} = 2\epsilon^* + 2(Z^* - Z)e^2\langle 1s'|\tfrac{1}{r}|1s'\rangle + e^2\langle 1s'(1)1s'(2)|\tfrac{1}{r_{12}}|1s'(1)1s'(2)\rangle$$
$$= 2\epsilon^* + 2(Z^* - Z)e^2\frac{Z^*}{a_0} + \frac{5}{8}\frac{Z^* e^2}{a_0}$$

last term was calculated earlier; first term is simply hydrogenic energy for two electrons, second term follows from expectation value of $\frac{1}{r}$ for hydrogenic atom

$$E_{\text{trial}} = \frac{e^2}{a_0}\left\{-Z^{*^2} + 2Z^*(Z^* - Z) + \frac{5}{8}Z^*\right\}$$

minimizing w.r.t. Z^* as a varying parameter, we find $\frac{\partial E_{\text{trial}}}{\partial Z^*} = 0$

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Value obtained using perturbation theory is -74.8 eV.